

An A.M.O.R.C. Note on Quasi-Infinite Graph Color Bags

Lean-verified graph constructions and a four-color search envelope

Abstract

We formalize a conservative reading of the phrase “quasi-infinite adjoint” for simple graphs. The useful construction is not a reflector into colorable graphs. Instead, it is an arbitrary indexed coproduct of graph factors, with countable edge unions and inverse-limit thread graphs as complementary infinite constructions. The Lean development proves the coproduct universal property, propagation of heterogeneous colorings through a sigma-sum palette, a universal property for countable edge unions, inherited colorings for inverse limits, and a small obstruction theorem showing why a naive free colorable reflector fails.

The addendum connects this machinery to quantum spatial search: the four-color theorem gives every finite planar graph a complete multipartite four-color completion. That completion is Laplacian integral, so the deterministic-search theorem of Li, Luo, Feng, and Li applies to the completion, though not automatically to the original planar graph.

Formal Setting

A simple graph on a vertex type V is represented by an adjacency predicate $\text{Adj} : V \rightarrow V \rightarrow \text{Prop}$ together with symmetry and irreflexivity. A graph homomorphism $G \rightarrow H$ is a vertex function preserving adjacency. A coloring by a color type C is a homomorphism into the complete graph K_C , where $x \text{ Adj } y$ means $x \neq y$.

The Lean file uses a local `Bijection` structure rather than importing a full category theory library. This keeps the artifact small while still stating the relevant adjoint-shaped equivalences explicitly.

The Bag Construction

Given an index type I , vertex families V_i , and graphs G_i on V_i , define

$$\text{vertices}(\text{SigmaGraph}(G)) = \sum_{i \in I} V_i.$$

Two tagged vertices are adjacent only when they have the same tag and the underlying vertices are adjacent in that factor:

$$(i, a) \text{ Adj } (j, b) \Leftrightarrow \exists k, i = k \wedge j = k \wedge a \text{ Adj}_{G_k} b.$$

Lean verifies the universal property:

$$\text{Hom}\left(\sum_{i \in I} G_i, H\right) \simeq \prod_{i \in I} \text{Hom}(G_i, H).$$

This is the most literal successful version of the “bag of factors” idea. It is arbitrary in the index type, so it can be finite, countable, or larger, depending on the ambient universe.

Heterogeneous Color Factors

If every factor has its own palette C_i and coloring

$$\text{color}_i : G_i \rightarrow K_{C_i},$$

then the whole bag is colored by the sigma-sum palette:

$$\sum_{i \in I} G_i \rightarrow K_{\sum_{i \in I} C_i}, \quad (i, a) \mapsto (i, \text{color}_{i(a)}).$$

If all factors share one palette C , the coproduct reuses that same palette. This captures the heterogeneous case from the motivating conversation without forcing all components to have the same chromatic number.

Countable Approximation

For graphs G_n on a shared vertex type V , Lean defines an edge-union graph:

$$x \text{ Adj } y \Leftrightarrow \exists n, x \text{ Adj}_{G_n} y.$$

A homomorphism out of this union is equivalent to one vertex map that preserves adjacency at every stage:

$$\text{Hom}(\cup_n G_n, H) \simeq \{f : V \rightarrow W \mid \forall n, f \text{ preserves Adj}_{G_n}\}.$$

This is a lightweight colimit-style theorem for staged approximants. It is a reasonable formal proxy for “take the finite constructions and let them run quasi-infinitely”, while still avoiding unverified analytic claims about metric or fractal dimension.

Inverse Threads

For an inverse system of vertex types

$$\dots \rightarrow V_2 \rightarrow V_1 \rightarrow V_0,$$

Lean defines a thread as a sequence $x_n \in V_n$ compatible with the bonding maps. The inverse-limit graph makes two threads adjacent when all finite coordinates are adjacent. Each coordinate projection is a graph homomorphism:

$$\pi_n : \lim G_n \rightarrow G_n.$$

Consequently, any coloring of one finite coordinate induces a coloring of the inverse-limit graph by composition with π_n .

The Failed Reflector

One tempting reading of “quasi-infinite adjoint” is: construct a free or largest C -colorable graph associated to any graph G . In the ordinary graph homomorphism category, that version fails before it gets started.

Lean proves:

$$\neg \text{Colorable}(G, C) \wedge \text{Colorable}(H, C) \Rightarrow \neg \exists (G \rightarrow H).$$

The reason is direct: if $G \rightarrow H$ and $H \rightarrow K_C$, then composition gives $G \rightarrow K_C$, contradicting non-colorability of G .

The development also formalizes the concrete witness: the triangle is not two-colorable, so there is no homomorphism from the triangle into any Bool-colorable graph. Thus an ordinary reflector with a unit $G \rightarrow L(G)$ into the full subcategory of two-colorable graphs cannot exist.

A.M.O.R.C. Addendum: Four-Color Search Envelope

The fetched spatial-search papers make the relevant graph invariant clear. In the Chakraborty–Novo–Ambainis–Omar CTQW model, the search Hamiltonian is

$$H_G = -P_w - \gamma A_G,$$

where A_G is the adjacency matrix. Their optimality criterion is spectral: the largest adjacency eigenvector should be the uniform state and the rest of the normalized spectrum should be bounded away from it. King–Linnebacher–Orth–Rizzi–Morigi use a Laplacian search Hamiltonian

$$H_\alpha = -\gamma_0 L_\alpha - P_w,$$

and identify the Laplacian spectral gap/algebraic connectivity as the critical resource for long-range tunneling search. Li–Luo–Feng–Li prove that any Laplacian integral graph admits deterministic search with certainty when the marked proportion is known in advance. Sadowski studies spatial search directly on planar Apollonian networks and finds that planar sparsity alone is not enough to make all marked-location cases operationally uniform.

This suggests a modest but real theorem.

Theorem (four-color Laplacian-integral envelope). Let $G = (V, E)$ be a finite loopless planar graph. Choose a four-coloring $c : V \rightarrow \{1, 2, 3, 4\}$, supplied by the four-color theorem. Define the completion K_c on the same vertex set by

$$x \text{Adj}_{K_c} y \Leftrightarrow c(x) \neq c(y).$$

Then:

- G is a spanning subgraph of K_c ;
- K_c is complete multipartite with at most four parts;
- K_c is Laplacian integral;
- therefore the deterministic quantum search theorem for Laplacian integral graphs applies to K_c .

Proof. If $x \text{Adj}_G y$, then properness of c gives $c(x) \neq c(y)$, so the identity on vertices is a graph homomorphism $G \rightarrow K_c$. The definition of K_c connects exactly the vertices in distinct color classes, hence K_c is complete multipartite.

Let the nonempty color classes have sizes $n_1, \dots, n_r$, with $r \leq 4$ and $N = n_1 + \dots + n_r$. For the complete multipartite graph $K_{n_1, \dots, n_r}$, a vector summing to zero inside part i is a Laplacian eigenvector with eigenvalue $N - n_i$, giving multiplicity $n_i - 1$. On the subspace of vectors constant on each part, the Laplacian acts by $a_i \mapsto N a_i - \sum_j n_j a_j$, whose eigenvalues are 0 once and N with multiplicity $r - 1$. Thus all Laplacian eigenvalues are integers:

$$\text{spec}(L(K_c)) = \{0, N, N - n_1, \dots, N - n_r\}.$$

So K_c is Laplacian integral. Li–Luo–Feng–Li’s deterministic-search theorem applies to every Laplacian integral graph, hence to K_c .

The Lean artifact verifies the graph-theoretic part of this theorem: `ColorCompletion` constructs K_c , `Coloring.toColorCompletion` proves the identity homomorphism $G \rightarrow K_c$, and the same-color/different-color lemmas prove the complete multipartite edge rule. The Laplacian spectrum calculation is ordinary linear algebra and is not yet mechanized in this repository.

Hamiltonian Reading

After ordering vertices by color classes, the adjacency matrix of a planar graph has zero diagonal color blocks:

$$A_G = \begin{pmatrix} 0 & A_{12} & A_{13} & A_{14} \\ A_{21} & 0 & A_{23} & A_{24} \\ A_{31} & A_{32} & 0 & A_{34} \\ A_{41} & A_{42} & A_{43} & 0 \end{pmatrix}.$$

The four-color completion replaces each off-diagonal block A_{ij} by an all-ones rectangular block. In CTQW language, it is a graph-engineering move: the planar Hamiltonian support is embedded into a four-sector long-range Hamiltonian support with complete cross-color tunneling. This connects the four-color theorem to spatial search without claiming that the original planar graph is Grover-optimal or deterministic-search-ready.

The same observation gives a boundary condition for the long-range tunneling paper: if all nonzero long-range couplings are treated as edges, then the support graph is complete for $N > 4$, hence not planar and not four-colorable. So four-color structure is a finite-range or thresholded-support invariant; it does not survive an all-positive long-range completion except as a chosen decomposition of the original planar skeleton.

What Was Actually Obtained

The verified construction is aggressive in size but conservative in claims:

- arbitrary indexed coproducts of graph factors;
- heterogeneous and shared-palette color propagation;
- a countable edge-union universal property;
- inverse-limit thread graphs with projections and inherited colorings;
- a formal obstruction to the naive colorable-reflector idea.
- a four-color completion theorem connecting planar graphs to Laplacian integral deterministic-search host graphs.

This gives a precise mathematical home for a large part of the phrase “quasi-infinite adjoint”: it is the coproduct/colimit/inverse-limit toolkit for colorable graph components. It does not yet construct graphs with a designed fractal dimension. To do that honestly, the next layer would need a metric or growth structure on the approximants, then a verified dimension invariant.

The A.M.O.R.C. addendum gives one concrete bridge to quantum spatial search: four-colorability provides a canonical class of Laplacian-integral completions for planar instances. The bridge is structural, not an optimality theorem for the original graph.

Lean Artifact

The formalization is in:

`Graphplay/Basic.lean`

It builds with Lean 4.30.0 using:

```
lake build
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References Fetched

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- Weisstein. Four-Color Theorem, MathWorld.